Answer Key, Quiz 2
HS 239, 2026

1. In autarky, Mr. X has wealth W_0 if there is no accident (probability p) and $W_0 - d$ if there is an accident (probability $1-p$). An insurance firm offers a contract (α, I), where α is premium and I is indemnity. The insurance is not fair in the sense that $\alpha = (1-p)mI$, where $m > 1$. You may assume that Mr. X has a concave Bernoulli utility function of wealth with the standard properties.

a) Mathematically (no graph in required) show that Mr. X, if he purchases insurance, will never purchase full insurance.

b) If the utility function is CARA (with β as the risk aversion parameter), how would the optimal α depend on W_0?

Ans

a) The consumer maximizes

$$\begin{aligned} & pu\left(W_0 - \alpha\right) + (1-p)\,u\left(W_0 - \alpha - d + I\right) \\ = \; & pu\left(W_0 - (1-p)mI\right) + (1-p)\,u\left(W_0 - (1-p)mI - d + I\right) \\ = \; & pu\left(W_0 - (1-p)mI\right) + (1-p)\,u\left(W_0 + (1-(1-p)\,m)\,I - d\right) \end{aligned}$$

Ideally the consumer chooses α, but, given the insurance line, he/she can choose I^*. Then we can backtrack and get $\alpha^* = m\,(1-p)\,I^*$.

Maximizing expected utility w.r.t. I and assuming interior solution

$$p\,(1-p)\,m * u'\left(W_1\right) = (1-p) * (1-(1-p)\,m) * u'\left(W_2\right)$$

where

$$\begin{aligned} W_1 &= W_0 - \alpha \\ W_1 &= W_0 - \alpha - d + I \end{aligned}$$

The above equation can be written as

$$\frac{u'\left(W_1\right)}{u'\left(W_2\right)} = \frac{1-m+pm}{pm} = \frac{pm+(1-m)}{pm}$$

Given $m > 1$, we have

$$\begin{aligned} \frac{pm+(1-m)}{pm} &< 1 \\ u'\left(W_1\right) &< u'\left(W_2\right) \end{aligned}$$

Since u is concave, $W_1 > W_2 \rightarrow$the consumer does not buy full insurance in equilibrium (no accident wealth > accident wealth).

b) Here, $u\left(W\right) = -e^{-\beta W} \rightarrow u'\left(W\right) = \beta e^{-\beta W}$

Putting the in first order condition above

$$\begin{aligned}\frac{e^{-\beta(W_0-\alpha)}}{e^{-\beta(W_0-\alpha-d+I)}} &= \frac{pm+(1-m)}{pm} = \Theta \text{ (a fraction)}\\ &\rightarrow e^{\beta(-d+I)} = \Theta\\ &\rightarrow \beta(I-d) = \ln(\Theta)\\ &\rightarrow I = \frac{1}{\beta}(\ln\Theta + d\beta)\end{aligned}$$

Since optimal I^* does not depend on W_0, optimal α^* does not depend on W_0 either.

MCQ

1. In the model of investment in risky asset using EU framework, a mean preserving contraction of interest rates will keep returns (**same**/increasing/decreasing) and risk (same/increasing/**decreasing**).

(2 points)

A mean preserving spread keep returns same and increases risk. Therefore, a mean preserving contraction (interest rates come closer) will keep the return same and decrease the risk.

2. In case of loaded insurance premium, it is sometimes possible to have full insurance. (True/False).

True. If the loading is additive, or if the utility function exhibits infinite risk aversion (L shaped indifference curves with the corner on the 45 degree line) in presence of multiplicative loading, the end result is on 45 degree line (full insurance)

3. In the model of investment in risky asset using EU framework, suppose I initially invest Rs 100 for risky asset. Then interest rate increases by 5% in both states of the world. Therefore, I will now invest Rs ______________ for risky asset.

Interest rates now are $\tilde{r}(1+.05) = 1.05 * \tilde{r}$. *Then new investment is*

$$a^* = \frac{100}{1.05} = 95.24$$

4. Consider the mean variance model of investment in one risky asset and one safe asset. Assume that the risky asset mean return is 1.9, while the safe asset return is 1.4. The risky asset has standard deviation of 1. Then the Sharpe ratio is ______________.

Ans. We are given $\mu_0 = 1.9; R = 1.4, \sigma_0 = 1$. *Then Sharpe ratio is*

$$\frac{\mu_0 - R}{\sigma_0} = \frac{1.9-1.4}{1} = .5$$

5. Mean variance framework is compatible with expected utility framework if the Bernoulli utility function exhibits_______ and the stochastic wealth follow _________ distribution.

(2 points)

CARA, Normal

6. Consider the mean variance model of investment in two risky assets under usual assumptions. Variance of asset 1 is 2 and that of asset 2 is 4. For diversification to work, we need the covariance is less than _________.

We have $\sigma_1 = \sqrt{2.0}$ *and* $\sigma_2 = 2$. *Hence, we need* $\rho < \min\left(\frac{\sigma_1}{\sigma_2}, \frac{\sigma_2}{\sigma_1}\right) = \min\left(\frac{1}{\sqrt{2}}, \sqrt{2}\right) = \frac{1}{\sqrt{2.0}} = 0.71$ *(approx)*

7. Consider the model of investment in risky asset using EU framework. Suppose a representative consumer in the economy has the Bernoulli utility function $U = \ln(Y)$. With economic growth, we expect ___________ (more/less/same) investment in risky asset in this economy.

We have $U' = \frac{1}{Y}$ *and* $U'' = -\frac{1}{Y^2} \rightarrow R_A(Y) = \frac{1}{Y}$, *which shows DARA. Hence more investment in risky asset can be expected in the economy as Y* increases.